

Consider the four data points

x	0	1	2	3
$f(x)$	1	2	9	28

Using the **Newton divided-difference algorithm**:

1. Construct the complete divided-difference table.
2. Write the degree-3 interpolating polynomial in **nested Newton form**.
3. Use your polynomial to evaluate $P_3(4)$.

Definition. Divided differences Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Let $x_0, x_1, \dots, x_n$ be $n + 1$ distinct points. Define $f[x_0, x_1, \dots, x_n]$.

Theorem. Recursive formula for divided differences Let $x_0, x_1, \dots, x_n$ be distinct points. Then

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}.$$

Let P_n be the unique polynomial of degree at most n that interpolates f at the $n + 1$ points:

$$x_0, x_1, \dots, x_n.$$

In Newton form,

$$P_n(x) = \underline{\hspace{15cm}} \quad (1)$$

The coefficient of x^n in P_n is

$$f[x_0, \dots, x_n].$$

Now let Q_{n-1} be the interpolation polynomial of degree at most $n - 1$ through the n data points:

$$(x_0, f(x_0)), \dots, (x_{n-1}, f(x_{n-1})),$$

The coefficient of x^{n-1} in Q_{n-1} is

$$\underline{\hspace{15cm}} \quad (2)$$

And let R_{n-1} be the interpolation polynomial of degree at most $n - 1$ through the n data points:

$$(x_1, f(x_1)), \dots, (x_n, f(x_n)).$$

The coefficient of x^{n-1} in R_{n-1} is

$$\underline{\hspace{15cm}} \quad (3)$$

Consider the polynomial

$$S_n(x) = \frac{(x - x_0)R_{n-1}(x) - (x - x_n)Q_{n-1}(x)}{x_n - x_0}.$$

We claim that $S_n = P_n$.

$$\text{First the degree of } S_n \text{ is } \underline{\hspace{15cm}}. \quad (4)$$

$$\text{because } \underline{\hspace{15cm}}. \quad (5)$$

Second, we claim that S_n interpolates f at the $n + 1$ points:

$$x_0, x_1, \dots, x_n.$$

Indeed, for $i = 1, \dots, n - 1$, we have

$$S_n(x_i) = \underline{\hspace{15cm}} \quad (6)$$

(7)

At the two endpoints,

$$S_n(x_0) = \text{_____} \quad (8)$$

$$\text{because } \text{_____} \quad (9)$$

and

$$S_n(x_0) = \text{_____} \quad (10)$$

$$\text{because } \text{_____} \quad (11)$$

Thus S_n interpolates f at all $n + 1$ points $x_0, \dots, x_n$.

S_n and P_n are two polynomials of degree at most n that interpolates f at the $n + 1$ points $x_0, \dots, x_n$, thus by uniqueness of the interpolating polynomial, we have

$$S_n = P_n.$$

It remains to compare the coefficients of x^n . The coefficient of x^n in

$$(x - x_0)R_{n-1}(x)$$

is

$$\text{_____} \quad (12)$$

While the coefficient of x^n in

$$(x - x_n)Q_{n-1}(x)$$

is

$$\text{_____} \quad (13)$$

Therefore the coefficient of x^n in S_n is

$$\text{_____} \quad (14)$$

But $S_n = P_n$, and the coefficient of x^n in P_n is $f[x_0, \dots, x_n]$. Consequently, our conclusion is

$$\text{_____} \quad (15)$$

Remainder theorem for polynomial interpolation. Let $x_0, x_1, \dots, x_n$ be distinct points, and let P_n be the polynomial of degree at most n that interpolates f at these points:

$$P_n(x_i) = f(x_i), \quad i = 0, 1, \dots, n.$$

Let x be any point in the domain of f . Assume that $f \in C^{n+1}(J)$, where J is an interval containing

$$x_0, x_1, \dots, x_n, x.$$

Then there exists a point $c \in J$, lying between the smallest and largest of

$$x_0, x_1, \dots, x_n, x,$$

such that

$$f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} \prod_{i=0}^n (x - x_i).$$

If $x = x_i$ for some i , then

$$f(x) - P_n(x) = 0$$

and

$$\prod_{i=0}^n (x - x_i) = 0,$$

so the result is immediate. Assume, therefore, that

$$\text{_____} \quad (16)$$

Now regard x as one additional interpolation point. By the Newton form of the interpolating polynomial through

$$x_0, x_1, \dots, x_n, x,$$

we have

$$f(x) = P_n(x) + \text{_____} \quad (17)$$

Define

$$h(t) = f(t) - P_n(t) - f[x_0, x_1, \dots, x_n, x] \prod_{i=0}^n (t - x_i).$$

For each $i = 0, 1, \dots, n$,

$$h(x_i) = \text{_____} \quad (18)$$

because

$$\text{_____} \quad (19)$$

Moreover, from the preceding Newton interpolation formula,

$$h(x) = \text{_____} \quad (20)$$

Thus h has

$$(\text{enter a number}) \text{_____} \quad (21)$$

distinct zeros:

$$(\text{give all zeros of } h \text{ that you know of}) \text{_____} \quad (22)$$

By repeated applications of Rolle's theorem, there exists a point c , lying between the smallest and largest of these points, such that

$$\text{_____} \quad (23)$$

Since P_n has degree at most n ,

$$P_n^{(n+1)}(t) = \text{_____} \quad (24)$$

Also,

$$\frac{d^{(n+1)}}{dt} \left(\prod_{i=0}^n (t - x_i) \right) = \text{_____} \quad (25)$$

Hence

$$h^{(n+1)}(t) = \text{_____} \quad (26)$$

Evaluating at $t = c$ gives

$$\text{_____} \quad (27)$$

$$\text{_____} \quad (28)$$

Substituting this into the error expression obtained above yields

$$\text{_____} \quad (29)$$